

Kunj Rathod

AI Engineer — Embodied AI, Robotics Systems & Low-Latency Inference

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EDUCATION

B.S. Computer Science — GPA 3.7/4.0, Dean's List

University of Utah

Aug 2023 – Dec 2026

Salt Lake City, UT

- Relevant: Machine Learning, Computer Vision, NLP, Distributed Systems, Algorithms, Multimodal LLM Agents

RESEARCH

Spatial Memory for Embodied & Long-Horizon Web Agents

University of Utah

2026 – Present

- Designing retrieval-augmented memory architectures (spatial, temporal, episodic, semantic) for long-horizon robot navigation and manipulation; inspired by ReMEmbR / NaVQA paradigm.
- Studying SceneSmith (MIT CSAIL / Toyota Research) for procedural generation of physics-ready simulation environments, enabling scalable robotic policy evaluation without manual scene authoring.

Long-term Active Embodied QA (LA-EQA)

University of Utah | w/ Nikesh Subedi

Spring 2026

- Co-authored “Exploring Long-term Active Embodied QA in Simulated Indoor Environments.” Proposed **Video Mind Palace (VMP)**: dropped Robotic Mind Palace’s expensive scene-graph pipeline in favor of direct VLM video queries, achieving **31–57% inference speedup** with only 13–17% relative accuracy cost on the full 100-question benchmark (A100, Qwen3-VL 8B/30B/FP8).
- Proposed benchmark evolution using SceneSmith + MuJoCo for interactive multi-agent, long-horizon embodied QA including object manipulation and intent attribution tasks.

Multi-Agent Materials Discovery — STARS Lab

University of Utah | NASA, Microsoft, U.S. DoD

Aug 2025 – Feb 2026

Salt Lake City, UT

- Graph-augmented multi-agent pipeline ingesting 1,000+ aerospace materials papers; constraint-based design region engine (temperature, creep, pressure) for automated Ashby plot generation and materials selection for rocket engines and hypersonics.

EXPERIENCE

Software Engineer Intern — Azure Data

Microsoft

May 2026 – Present

Redmond, WA

- Distributed data systems engineering on Azure; full-stack development within the Azure Data platform.

AI Services Intern — SUDO Program

University of Utah Health

Jan 2025 – Present

Salt Lake City, UT

- **Inference pipeline:** p95 TTFT <200ms on AWS Bedrock with 40% latency reduction via pipeline optimization, API caching, DynamoDB+S3 session persistence for 1,000+ concurrent LLM conversations.
- Built event-driven Lambda orchestration (React/TypeScript frontend, Flask middleware) for a HIPAA-compliant platform serving 90+ hospital executives; shipped 6 features in 4 sprints.

AI Engineering Intern

CourtEasy.ai / Nugen

Nov 2024 – Apr 2025

Remote

- Production hybrid RAG over 10M+ legal documents at 5,000+ daily queries; +28% retrieval accuracy, -35% hallucination rate.
- Benchmarked 8 LLM families (F1, latency, cost) across 4 legal benchmarks; routing decisions saved \$50k+/yr projected inference spend.

AI Research Intern — BioGraphRAG

Garje Marathi Global

May 2024 – Aug 2024

Salt Lake City, UT

- Distributed GraphRAG over 1M+ biomedical entities on NebulaGraph; 3× graph traversal speedup to sub-500ms p95 via caching and high-degree node pruning; 2M+ entity ETL updates/month.

SELECTED PROJECTS

FNDR — Local AI Memory Engine (macOS)

Rust, Tauri, Metal/ONNX, Llama 3.2, Whisper, Apple Vision, MCP

2024 – Present

- **Edge AI:** Metal-accelerated on-device LLM (Llama 3.2) and VLM (SmolVLM) inference on Apple Silicon; real-time Apple Vision Framework OCR pipeline with CLIP visual embeddings for temporal scene reconstruction.
- Graphiti-style temporal knowledge graph for multi-hop reasoning across screen, audio, and web activity; local Whisper transcription; MCP server for agent interop. Zero cloud, zero telemetry.

FlowVia — V2X Urban Mobility Optimization System

Apr 2024

Python, TensorFlow, LSTM, V2X (DSRC/C-V2X), OBD-II

- Vehicle-to-Everything traffic platform: real-time speed recommendations via live SPaT data, LSTM traffic flow prediction (continuously retrained), AES-256 edge-encrypted V2V/V2I/V2N communication.
- Full system: in-vehicle OBD-II unit, mobile interface, cloud ML backend, roadside V2X unit — ultra-low-latency, edge-first architecture.

HirePilot — 9-Agent Recruiting System [Podium AI Hackathon — 1st Place] 2026
TypeScript, Anthropic API, Twilio, Google Calendar, Slack

- Autonomous multi-agent pipeline: GitHub sourcing, live voice AI interviews (Twilio + transcript analysis), multi-dimensional scoring, automated calendar + Slack approval workflows.

WRITING & PUBLICATIONS

- **Exploring LA-EQA in Simulated Indoor Environments** — Rathod, Subedi. Class report, CS Advanced AI, U of Utah, Spring 2026.
- **BioGraphRAG** — co-authored; republished by NebulaGraph. [Substack](#), Oct 2024.
- **FlowVia: Technical Deep Dive** — [Substack](#), Apr 2024.

TECHNICAL SKILLS

Languages	Python, C++, Rust, Java, TypeScript, SQL
AI / ML	PyTorch, TensorFlow, ONNX, RAG, GraphRAG, LangChain, LlamaIndex, Transformers, Multi-Agent Systems, Embodied AI, VLMs, RL (DQN, PPO)
Systems	Metal (Apple Silicon), Edge Inference, V2X Protocols (DSRC/C-V2X), AWS Bedrock, Lambda, DynamoDB, Docker, Kubernetes, NebulaGraph

AWARDS

Kahlert Impact Prize (\$1,000) — Kahlert School of Computing, U of Utah, Mar 2026 · **Dean’s List** 2024–25 · **Minute0 Hackathon Winner** Feb 2026